

HODGE THEORY VIA THE HEAT KERNEL

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1. DIFFERENTIAL FORMS ON RIEMANNIAN MANIFOLDS

Suppose (X, g) is a compact, orientable Riemannian manifold. Let $\Omega_X^k := \wedge^k T^*X$ be the k th exterior power of the cotangent bundle T^*X , and $A^k(X) := \Gamma(X, \Omega_X^k)$ be the real vector space of smooth k -forms on X . The Riemannian metric g induces a metric $\langle, \rangle$ on the vector bundle Ω_X^k , in such a way that if $e_1, \dots, e_n$ is an orthonormal basis for the tangent space $(T_x X, g_x)$, and $e_1^*, \dots, e_n^*$ is the dual basis, then the k -forms $e_{i_1} \wedge \dots \wedge e_{i_k}$ form an orthonormal basis for $\Omega_{X,x}^k$. Observe that if $\alpha, \beta \in A^k(X)$ are given k -forms, then $x \mapsto \langle \alpha, \beta \rangle_x$ is a smooth function on X .

Let $n = \dim X$. The Riemannian volume form on (X, g) , denoted vol , is the unique n -form on X such that in any oriented system of local coordinates $(x^1, \dots, x^n)$, we have

$$\text{vol} = \sqrt{\det(g_{ij})} dx^1 \wedge \dots \wedge dx^n,$$

where g_{ij} are the components of the metric in these coordinates. The Riemannian volume form allows us to integrate functions on X in a canonical way. In particular, the L^2 metric on the space of differential forms $A^k(X)$ is defined

$$(\alpha, \beta)_{L^2} := \int_X \langle \alpha, \beta \rangle \text{vol},$$

where we are integrating the smooth function $x \mapsto \langle \alpha, \beta \rangle_x$. The *Hodge operator* $*$: $A^k(X) \rightarrow A^{n-k}(X)$ is defined by the relation

$$(\alpha, \beta) \cdot \text{vol} = \alpha \wedge * \beta$$

for all $\alpha, \beta \in A^k(X)$. This determines $*$, since for each $x \in X$, the bilinear pairing

$$\wedge : \Omega_{X,x}^k \times \Omega_{X,x}^{n-k} \longrightarrow \Omega_{X,x}^n = \mathbb{R} \cdot \text{vol}_x$$

is nondegenerate. Note that $(\alpha, \beta)_{L^2} = \int_X \alpha \wedge * \beta$. Moreover, $*^2 = (-1)^{k(n-k)}$ on $A^k(X)$, and if $k = 0$, then $\text{vol} = *1$.

The exterior derivative d on the de Rham complex $A^\bullet(X)$ has a formal adjoint for the L^2 metric $d^* : A^k(X) \rightarrow A^{k-1}(X)$, which is defined

$$d^* := (-1)^k *^{-1} \circ d \circ *.$$

One checks that $(d\alpha, \beta)_{L^2} = (\alpha, d^* \beta)_{L^2}$ using Stokes' formula. The *Laplacian* is defined

$$\Delta := d^* d + d d^*.$$

Forms in the kernel of Δ are called *harmonic forms*.

2. THE HEAT EQUATION

Consider the bundle $\Omega_X^k \boxtimes \Omega_X^k$ on the product manifold $\mathbb{R}_{>0} \times X \times X$, together with the Laplacian $\Delta = \Delta_k$ on $A^k(X)$.

Definition 2.1. A *heat kernel* (for the Laplacian Δ_k) is a smooth section $p_t(x, y)$ of $\Omega_X^k \boxtimes \Omega_X^k$ such that:

- (1) $p_t(x, y)$ satisfies the equation

$$(\partial_t + \Delta_x)p_t(x, y) = 0;$$

- (2) we have $\lim_{t \rightarrow 0} P_t(\alpha) = \alpha$ for all $\alpha \in A^k(X)$, where for each $t > 0$, the *heat operator* $P_t : A^k(X) \rightarrow A^k(X)$ is defined by the formula

$$P_t(\alpha)(x) := \int_X \langle p_t(x, y), \alpha(y) \rangle_y dy.$$

The heat operators P_t satisfy the *heat equation*

$$(\partial_t + \Delta)P_t = 0. \tag{2.1}$$

The proof that the heat kernel exists can be found in these references...

Lemma 2.2. A heat kernel for the Laplacian Δ_k is unique if it exists.

Proof. Suppose p_t and p'_t are heat kernels, with corresponding heat operators P_t and P'_t . Then the difference $s_t := p_t - p'_t$ satisfies the heat equation for Δ_k , and we have $\lim_{t \rightarrow 0} S_t(\alpha) = 0$ for all $\alpha \in A^k(X)$, where $S_t = P_t - P'_t$.

Fix $t > 0$ and $\alpha, \beta \in A^k(X)$, and consider the function

$$f(\theta) := (S_\theta(\alpha), P_{t-\theta}(\beta)),$$

where $0 < \theta < t$. Using (2.1) and the fact that Δ is self-adjoint, we find that

$$\begin{aligned} f'(\theta) &= (\partial_\theta S_\theta(\alpha), P_{t-\theta}(\alpha)) + (S_\theta(\alpha), \partial_\theta P_{t-\theta}(\beta)) \\ &= (\partial_\theta S_\theta(\alpha), P_{t-\theta}(\beta)) + (S_\theta(\alpha), \Delta P_{t-\theta}(\beta)) \\ &= (\partial_\theta S_\theta(\alpha), P_{t-\theta}(\beta)) + (\Delta S_\theta(\alpha), P_{t-\theta}(\beta)) \\ &= ((\partial_\theta + \Delta)S_\theta(\alpha), P_{t-\theta}(\beta)) \\ &= 0 \end{aligned}$$

so $f(\theta)$ is a constant function. Since $\lim_{\theta \rightarrow 0} S_\theta(\alpha) = 0$, we have $\lim_{\theta \rightarrow 0} f(\theta) = 0$, which implies $f(\theta) = 0$. So we get that

$$(S_t(\alpha), \beta) = \lim_{\theta \rightarrow t} f(\theta) = 0.$$

Since β was arbitrary, we must have $S_t(\alpha) = 0$. This implies that for each fixed $x \in X$, we have $\langle s_t(x, y), \alpha(y) \rangle_y = 0$ for all $y \in X$, and so we finally conclude that $s_t(x, y) = 0$. $\square$

Proposition 2.3. The heat operators P_t are self-adjoint.

Proof. Fix $t > 0$ and $\alpha, \beta \in A^k(X)$, and define a function

$$f(\theta) := (P_\theta(\alpha), P_{t-\theta}(\beta))$$

for $0 < \theta < t$. Proceeding as in the proof of Lemma 2.2, we see that $f(\theta)$ is constant. Considering the limits as $\theta \rightarrow 0$ and $\theta \rightarrow t$ then shows that $(\alpha, P_t(\beta)) = (P_t(\alpha), \beta)$, as required. $\square$

Proposition 2.4. *The heat operators P_t form a semigroup: we have $P_{t_2}P_{t_1} = P_{t_1+t_2}$ for all $t_1, t_2 > 0$.*

Proof. Suppose $0 < s < t$, and $\alpha \in A^k(X)$. Then

$$\begin{aligned} P_{t-s}P_s(\alpha)(x) &= P_{t-s} \int_X \langle p_s(x, y), \alpha(y) \rangle_y dy \\ &= \int_X \left\langle p_{t-s}(x, z), \int_X \langle p_s(z, y), \alpha(y) \rangle_y dy \right\rangle_z dz \\ &= \int_{X \times X} \langle p_{t-s}(x, z), \langle p_s(z, y), \alpha(y) \rangle_y \rangle_z dy dz \\ &= \int_{X \times X} \langle \langle p_{t-s}(x, z), p_s(z, y) \rangle_z, \alpha(y) \rangle_y dz dy \\ &= \int_X \left\langle \int_X \langle p_{t-s}(x, z), p_s(z, y) \rangle_z dz, \alpha(y) \right\rangle_y dy, \end{aligned}$$

so $P_{t-s}P_s$ has integral kernel $\int \langle p_{t-s}(x, z), p_s(z, y) \rangle_z dz$. We claim that this kernel is equal to p_t . Differentiating under the integral sign, we find that

$$(\partial_t + \Delta_x) \int \langle p_{t-s}(x, z), p_s(z, y) \rangle_z dz = 0.$$

Moreover, if $\alpha \in A^k(X)$, then

$$\begin{aligned} \lim_{s \rightarrow 0} P_{t-s}P_s(\alpha) &= \lim_{s \rightarrow 0} P_{t-s} \left(\lim_{s \rightarrow 0} P_s(\alpha) \right) \\ &= \lim_{s \rightarrow 0} P_{t-s}(\alpha) \\ &= P_t(\alpha), \end{aligned}$$

so we have

$$\lim_{t \rightarrow 0} P_{t-s}P_s(\alpha) = \lim_{t \rightarrow 0} \lim_{s \rightarrow 0} P_{t-s}P_s(\alpha) = \lim_{t \rightarrow 0} P_t(\alpha) = \alpha.$$

Therefore, $\int \langle p_{t-s}(x, z), p_s(z, y) \rangle_z dz$ satisfies the two defining properties of the heat kernel, so it is equal to p_t by Lemma 2.2. $\square$

We now consider the extensions of P_t and p_t to the Hilbert space completions of $A^k(X)$ and $A^k(X)^{\otimes 2}$, which we denote by $A_{L^2}^k(X)$ and $A_{L^2}^k(X)^{\otimes 2}$. Note that the inner product on $A^k(X)^{\otimes 2}$ is defined on simple tensors by

$$(\omega_1 \otimes \eta_1, \omega_2 \otimes \eta_2)_{L^2} := (\omega_1, \omega_2) \cdot (\eta_1, \eta_2),$$

and is then extended linearly. If $t > 0$ is fixed, then $p_t(x, y) \in A^k(X)^{\otimes 2}$, so it can be written

$$p_t(x, y) = \sum_i p_i(x) \otimes q_i(y)$$

for certain smooth k -forms p_i, q_i , whose dependence on t we suppress. This expression is not in any way canonical, but we will find it helpful for establishing basic facts about the heat operators P_t . Note that the definition implies that

$$\|p_t\|^2 = \sum_{i,j} (p_i, p_j)(q_i, q_j). \quad (2.2)$$

Lemma 2.5. *For each fixed $t > 0$, the double form p_t is square-integrable.*

Proof. This follows immediately from (2.2) and the fact that X is compact. $\square$

Lemma 2.6. *The heat operators P_t are continuous linear operators, and so extend continuously to operators on $A_{L^2}^k(X)$.*

Proof. Suppose $(\alpha_n)_{n \geq 0}$ is a sequence in $A^k(X)$ which converges to zero in the L^2 norm. We have to show that $\lim_{n \rightarrow \infty} \|P_t(\alpha_n)\| = 0$. If $x \in X$, then we have

$$\begin{aligned} P_t(\alpha_n)(x) &= \int_X \langle p_t(x, y), \alpha_n(y) \rangle_y dy \\ &= \sum_i \left(\int_X \langle q_i, \alpha_n \rangle \text{vol} \right) p_i(x) \\ &= \sum_i (q_i, \alpha_n) p_i(x). \end{aligned}$$

Since $q_i \in A_{L^2}^k(X)$, the Cauchy-Schwarz inequality implies

$$|(q_i, \alpha_n)| \leq \|q_i\| \|\alpha_n\|,$$

so we have

$$\begin{aligned} \|P_t(\alpha_n)\|^2 &= \int_X \langle P_t(\alpha_n)(x), P_t(\alpha_n)(x) \rangle_x dx \\ &= \sum_{i,j} (q_i, \alpha_n)(q_j, \alpha_n) \int_X \langle p_i(x), p_j(x) \rangle_x dx \\ &\leq \sum_{i,j} |(q_i, \alpha_n)| |(q_j, \alpha_n)| |(p_i, p_j)| \\ &\leq \left(\sum_{i,j} \|p_i\| \|q_i\| \|p_j\| \|q_j\| \right) \|\alpha_n\|^2. \end{aligned}$$

This shows $\lim_{n \rightarrow \infty} \|P_t(\alpha_n)\| = 0$ as needed. $\square$

Proposition 2.7. *The heat operators P_t are compact.*

Proof. We show that P_t can be written as the limit, in the operator norm, of a sequence of operators with finite dimensional image. It follows easily from this that P_t is compact, in the sense that the image under P_t of every bounded sequence has a convergent subsequence.

Suppose $(e_i)_{i \geq 0}$ is an orthonormal basis for $A_{L^2}^k(X)$. Since $A^k(X)$ is dense in $A_{L^2}^k(X)$, we may assume that the e_i are smooth sections. It is immediate from the definition that $(e_i \otimes e_j)_{i,j \geq 0}$ is an orthonormal basis for $A_{L^2}^k(X)^{\otimes 2}$. By Lemma 2.5, we can write

$$p_t = \sum_{i,j=0}^{\infty} (p_t, e_i \otimes e_j) e_i \otimes e_j.$$

If we define

$$p_t^N := \sum_{i,j=0}^N (p_t, e_i \otimes e_j) e_i \otimes e_j,$$

then we have $p_t^N \rightarrow p_t$ as $N \rightarrow \infty$ in the L^2 norm. Let P_t^N be the operator on $A_{L^2}^k(X)$ defined by

$$P_t^N(\alpha)(x) := \int_X \langle p_t^N(x, y), \alpha(y) \rangle_y dy.$$

Since p_t^N is a finite sum of smooth sections, it is itself a smooth section, and an argument similar to Lemma 2.6 shows that P_t^N is a continuous operator. We have

$$\begin{aligned} P_t^N(\alpha(x)) &= \int_X \langle p_t^N(x, y), \alpha(y) \rangle_y dy \\ &= \sum_{i,j=1}^N (p_t, e_i \otimes e_j)(e_j, \alpha) e_i(x), \end{aligned}$$

for all $\alpha \in A_{L^2}^k(X)$, so the image of P_t^N is contained in the span of $e_1, \dots, e_N$. Finally, if $\alpha \in A_{L^2}^k(X)$ is such that $\|\alpha\| \leq 1$, then the same calculation shows that

$$\begin{aligned} \|P_t(\alpha) - P_t^N(\alpha)\|^2 &= \sum_{i,j=N+1}^{\infty} |(p_t, e_i \otimes e_j)|^2 |(e_j, \alpha)|^2 \\ &\leq \sum_{i,j=N+1}^{\infty} |(p_t, e_i \otimes e_j)|^2 = \|p_t - p_t^N\|^2, \end{aligned}$$

and so we have $\|P_t - P_t^N\| \leq \|p_t - p_t^N\| \rightarrow 0$ as $N \rightarrow \infty$. $\square$

3. THE SPECTRUM OF THE LAPLACIAN

The functional analysis of the heat operators P_t is summarised in the following theorem:

Theorem 3.1. *The Hilbert space $A_{L^2}^k(X)$ has an orthonormal basis $(\varphi_i)_{i \geq 1}$ of simultaneous eigenvectors for the heat operators P_t , where the φ_i are smooth k -forms. The eigenvalues for P_t are of the form $e^{-\lambda_i t}$, where the λ_i are real numbers satisfying*

$$0 \leq \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \dots$$

The sequence $(\lambda_i)_{i \geq 1}$ accumulates only at infinity; in particular, each λ_i occurs with only finite multiplicity. Finally, we have

$$\Delta \varphi_i = \lambda_i \varphi_i$$

for all $i \geq 1$, and if $\varphi \in A^k(X)$ is an eigenvector for Δ with eigenvalue $\lambda \in \mathbb{R}$, then $\lambda = \lambda_i$ for some i , and φ is a simultaneous λ_i -eigenvector for the P_t .

Proof. By Propositions 2.3, 2.4, and 2.7, the heat operators P_t form a commuting family of compact self-adjoint operators on $A_{L^2}^k(X)$, so the existence of $(\varphi_i)_{i \geq 1}$ follows from the spectral theorem for compact self-adjoint operators. Note that the corresponding eigenvalues $\gamma_i(t) \in \mathbb{R}$ are smooth functions of t , because the heat kernel p_t is smooth in t .

We begin by showing $\gamma_i(t) > 0$ for all $t > 0$. By Propositions 2.3 and 2.4, we have

$$\begin{aligned} \gamma_i(t) \|\varphi_i\|^2 &= (P_t(\varphi), \varphi) \\ &= (P_{t/2} P_{t/2}(\varphi), \varphi) \\ &= (P_{t/2}(\varphi), P_{t/2}(\varphi)) \geq 0, \end{aligned}$$

so we at least have $\gamma_i(t) \geq 0$. If $\gamma_i(t) = 0$ for some t , then the same calculation shows $P_{t/2}(\varphi_i) = 0$. Continuing like this, we see that $P_s(\varphi_i) = 0$ for $s = \frac{t}{2^n}, n \geq 0$. Hence,

$$\varphi = \lim_{s \rightarrow 0} P_s(\varphi) = 0,$$

a contradiction.

It now follows that the eigenvectors φ_i are smooth, since $\varphi_i = \gamma_i(t)^{-1}P_t(\varphi)$, and the integral operator P_t takes values in smooth forms.

To further analyse the $\gamma_i(t)$, note that the heat equation implies

$$\begin{aligned} 0 &= (\partial_t + \Delta)P_t(\varphi_i) \\ &= (\partial_t + \Delta)\gamma_i(t)\varphi_i \\ &= \gamma_i'(t)\varphi_i + \gamma_i(t)\Delta\varphi_i, \end{aligned}$$

and since $\gamma_i(t) \neq 0$, we have

$$\Delta\varphi_i = -\frac{\gamma_i'(t)}{\gamma_i(t)}\varphi_i. \quad (3.1)$$

But $\Delta\varphi_i$ is independent of t , so $\lambda_i := -\frac{\gamma_i'(t)}{\gamma_i(t)}$ is a constant. Solving the differential equation $\gamma_i'(t) = -\lambda_i\gamma_i(t)$, we find that $\gamma_i(t) = Ce^{-\lambda_i t}$ for some $C \in \mathbb{R}$. Now

$$\varphi_i = \lim_{t \rightarrow 0} P_t(\varphi_i) = \lim_{t \rightarrow 0} Ce^{-\lambda_i t}\varphi_i = C\varphi_i,$$

so $C = 1$, and the eigenvalues $\gamma_i(t) = e^{-\lambda_i t}$ are of the required form.

From (3.1), we see that $\Delta\varphi_i = \lambda_i\varphi_i$. Moreover,

$$\begin{aligned} \lambda_i \|\varphi_i\|^2 &= (\Delta\varphi_i, \varphi_i) \\ &= (d^*d\varphi_i, \varphi_i) + (dd^*\varphi_i, \varphi_i) \\ &= \|d\varphi_i\|^2 + \|d^*\varphi_i\|^2 \geq 0, \end{aligned}$$

so $\lambda_i \geq 0$. Reordering the basis, we may assume the sequence $(\lambda_i)_{i \geq 1}$ is increasing:

$$0 \leq \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \dots$$

The fact that the only accumulation point of this sequence is at ∞ is a part of the spectral theorem.

Finally, suppose $\varphi \in A^k(X)$ is an eigenvector for Δ with eigenvalue λ . We have

$$\begin{aligned} P_t(\lambda\varphi) &= \Delta P_t(\varphi) \\ &= -\partial_t P_t(\varphi) \\ &= -\partial_t \sum_{i \geq 1} (\varphi, \varphi_i) e^{-\lambda_i t} \varphi_i \\ &= \sum_{i \geq 1} (\varphi, \varphi_i) \lambda_i e^{-\lambda_i t} \varphi_i. \end{aligned}$$

Taking the limit as $t \rightarrow 0$ shows that

$$\sum_{i \geq 1} (\varphi, \varphi_i) \lambda_i \varphi_i = \lambda\varphi = \sum_{i \geq 1} (\varphi, \varphi_i) \lambda_i \varphi_i;$$

hence, $(\varphi, \varphi_i)\lambda = (\varphi, \varphi_i)\lambda_i$ for all $i \geq 1$. Since $\varphi \neq 0$, we must have $(\varphi, \varphi_i) \neq 0$ for some i , which implies $\lambda = \lambda_i$. $\square$

By Theorem 3.1, the space $\mathcal{H} := \text{Ker}(\Delta)$ of harmonic forms is a finite dimensional subspace of $A^k(X)$. Since $\mathcal{H}$ is finite dimensional, it is closed in $A_{L^2}^k(X)$, and we have an orthogonal direct sum decomposition:

$$A_{L^2}^k(X) = \mathcal{H} \oplus \mathcal{H}^\perp. \quad (3.2)$$

Our task in the remainder of this section is to show that this decomposition exists on the level of smooth forms.

Lemma 3.2. *For all $\alpha \in \mathcal{H}^\perp \subset A_{L^2}^k(X)$, the integral*

$$\tilde{G}(\alpha) := \int_0^\infty P_t(\alpha) dt$$

is absolutely convergent, and defines a bounded linear map $\tilde{G} : \mathcal{H}^\perp \rightarrow \mathcal{H}^\perp \cap A^k(X)$. Moreover, on the subspace $\mathcal{H}^\perp \cap A^k(X)$, we have $\tilde{G} = \Delta^{-1}$.

Proof. Let $\lambda > 0$ be the smallest positive eigenvalue of Δ . For each $t > 0$, the operator norm of $(P_t)|_{\mathcal{H}^\perp}$ is equal to its largest eigenvalue $e^{-\lambda t}$, so we have

$$\left\| \int_0^\infty P_t dt \right\| \leq \int_0^\infty \|P_t\| dt = \int_0^\infty e^{-\lambda t} dt = \frac{1}{\lambda}.$$

This shows that $\tilde{G}$ is a bounded linear map $\mathcal{H}^\perp \rightarrow A^k(X)$.

It is clear that Δ preserves $\mathcal{H}^\perp$, since if $\alpha \in \mathcal{H}^\perp \cap A^k(X)$, then for all $\beta \in \mathcal{H}$, we have

$$(\beta, \Delta(\alpha)) = (\Delta(\beta), \alpha) = 0$$

by self-adjointness. Now if $\varphi_i \in \mathcal{H}^\perp$ is a eigenvector corresponding to the positive part of the spectrum of Δ , we have

$$\begin{aligned} \tilde{G}(\varphi_i) &= \int_0^\infty e^{-\lambda_i t} \varphi_i dt \\ &= \left(\int_0^\infty e^{-\lambda_i t} dt \right) \varphi_i \\ &= \left[-\frac{1}{\lambda_i} e^{-\lambda_i t} \right]_{t=0}^{t=\infty} \varphi_i \\ &= \frac{1}{\lambda_i} \varphi_i. \end{aligned}$$

This shows that $\tilde{G}$ preserves $\mathcal{H}^\perp$; moreover, since

$$\tilde{G} \circ \Delta(\varphi_i) = \varphi_i = \Delta \circ \tilde{G}(\varphi_i)$$

for all $\varphi_i \in \mathcal{H}^\perp$, the operators $\tilde{G} \circ \Delta$ and $\Delta \circ \tilde{G}$ extend continuously to the identity on $\mathcal{H}^\perp$. Therefore, $\tilde{G} = \Delta^{-1}$ on $\mathcal{H}^\perp \cap A^k(X)$. $\square$

Definition 3.3. The *Green's operator* $G : A_{L^2}^k(X) \rightarrow A^k(X)$ is defined to be the projection of $A_{L^2}^k(X)$ onto $\mathcal{H}^\perp$, followed by $\tilde{G}$.

Hence, by Lemma 3.2, the projection $A_{L^2}^k(X) \rightarrow \mathcal{H}^\perp$ is given by $\Delta \circ G$, and if $\alpha \in A^k(X)$ is smooth, then so is $\Delta G(\alpha)$. This implies that the decomposition (3.2) restricts to smooth forms, where it is known as the *Hodge decomposition*:

Theorem 3.4. *Let $H : A^k(X) \rightarrow \mathcal{H}$ denote the harmonic projection operator. Then each smooth k -form $\alpha \in A^k(X)$ can be written uniquely as a sum*

$$\alpha = H(\alpha) + \Delta G(\alpha),$$

which expresses an orthogonal direct sum decomposition:

$$A^k(X) = \mathcal{H}^k \oplus \text{Im}(d^{k-1}) \oplus \text{Im}((d^*)^{k+1}).$$